Enabling fluorescence capabilities in the quantitative oblique back-illumination microscopy (qOBM) handheld probe

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OVERVIEW

- Gliomas are brain tumors that develop in glial cells. Survival rates for patients range from 5-95%, influenced by tumor type, grade, size, and location.
- Treatment for these tumors involves tumor resection by neurosurgeons. Accurate diagnosis and full tumor resection are critical in improving survival.
- Unlike other tumors, brain tumors are excised without extra tissue margins to avoid damaging normal brain function.
- A widely used approach for tumor identification intraoperatively is **5-ALA**, an effective fluorescent marker for glioblastomas.
- However, this marker does not provide any diagnostic information about the tissue or detects low grade tumors.
- There is a lack of effective intraoperative tools available for real-time tissue diagnosis to guide surgical resection.

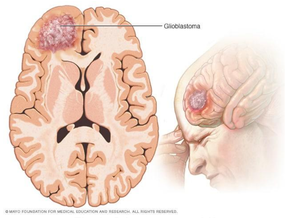


Fig 1: Glioblastoma [4]

Fig 2: CT Of Glioblastoma [1]

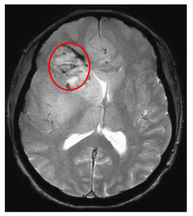
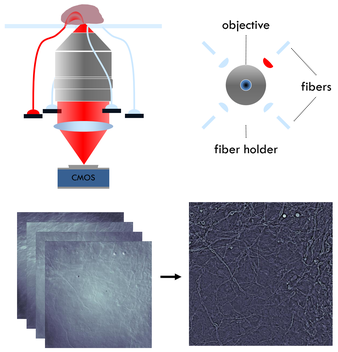


Fig3: MRI of Glioblastoma [1]

Quantitative back-illumination microscopy (qOBM) is a **label-free, real-time imaging modality that provides subcellular resolution of thick, scattering samples**, such as the brain. Here, we introduce qOBM as a handheld probe with added fluorescence capability. The combination of qOBM's label-free approach and fluorescence imaging serves as a promising technology for identifying tumor margins intraoperatively.

MATERIALS and METHODS

qOBM system



- Fiber based system
- Forward-scattered light is collected
- 4 images are captured and then combined

qOBM handheld probe

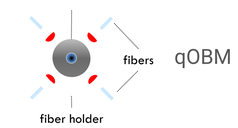
- Provides cellular and subcellular details and structures without labels
- Provides nuclear contrast in thick tissue with a handheld system that can be maneuvered around the sample
- Fluorescence imaging will be useful when imaging with fluorescent dyes like 5-ALA
- Could improve survival outcomes for glioma patients by allowing for better excision of the tumor

- **Fabrication of PCBs:**
  - Circuit design
  - Autodesk Eagle
- **Circuit Assembly:**
  - Soldering
- **Programming:**
  - Arduino
  - MATLAB
- **Probe Redesign:**
  - SOLIDWORKS
  - Implement longpass filter
  - 8 openings for fibers
- **Fluorescence Imaging:**
  - Non-biological and biological samples

RESULTS

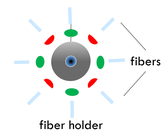
qOBM probe setups

Single channel



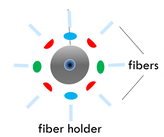
qOBM

Dual channel



qOBM + single channel fluorescence

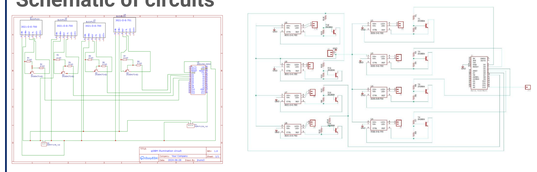
3 channel



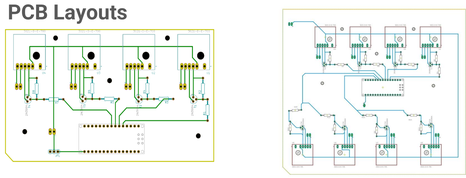
qOBM + Dual channel fluorescence

Fabrication of PCBs

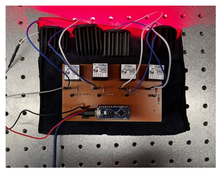
Schematic of circuits



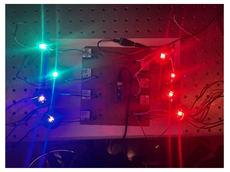
PCB Layouts



PCB Setups

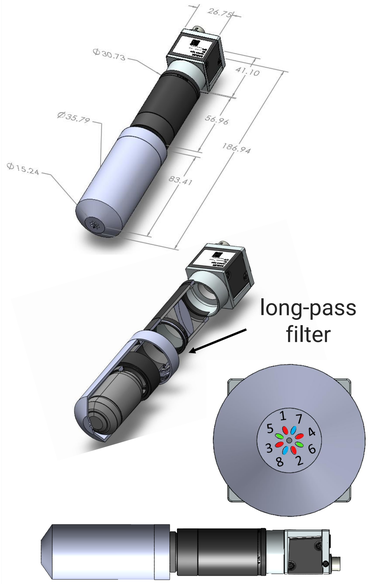


4-LED setup



8-LED setup

Handheld probe prototype



DISCUSSION and FUTURE WORK

We have presented the combined qOBM-fluorescence handheld probe, a label-free, real-time optical imaging modality that will help neurosurgeons precisely identify tumor margins intraoperatively. Eventually, this work will improve survival outcomes for glioma patients by allowing for improved tumor resection. Furthermore, this probe could be used to image a wide range of clinical and biological samples. Future work includes streamlining the portability of the probe by enabling full acquisition functionality with a laptop. Both biological and non-biological samples will also be imaged to test and refine the fluorescence capabilities of the probe once it is constructed.

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